MiMo-V2-Flash Technical Report

LLM-Core Xiaomi | arXiv:2601.02780v2 [cs.CL] 8 Jan 2026

ONE-LINE TAKEAWAY

MiMo-V2-Flash achieves strong reasoning and agentic performance with **309B** total parameters and **15B** active parameters, rivaling top-tier models like DeepSeek-V3.2 and Kimi-K2 while using only **1/2** and **1/3** of their parameters, respectively.

PROBLEM

- Long-context modeling must be simultaneously fast and strong to support advanced reasoning and autonomous agents.
- Scaling reinforcement learning (RL) for agent training is computationally expensive and inefficient.

METHOD

- Uses a Mixture-of-Experts (MoE) architecture with 256 total experts and 8 activated per token.
- Employs a hybrid attention mechanism combining Sliding Window Attention (SWA) with global attention at a 5:1 ratio.
- Utilizes a 128-token sliding window size with learnable attention sink bias to maintain long-context performance.
- Incorporates Lightweight Multi-Token Prediction (MTP) as a draft model for speculative decoding.
- Introduces Multi-Teacher On-Policy Distillation (MOPD) for efficient post-training with domain-specialized teachers.

FIGURE 1. Benchmark performance of MiMo-V2-Flash.

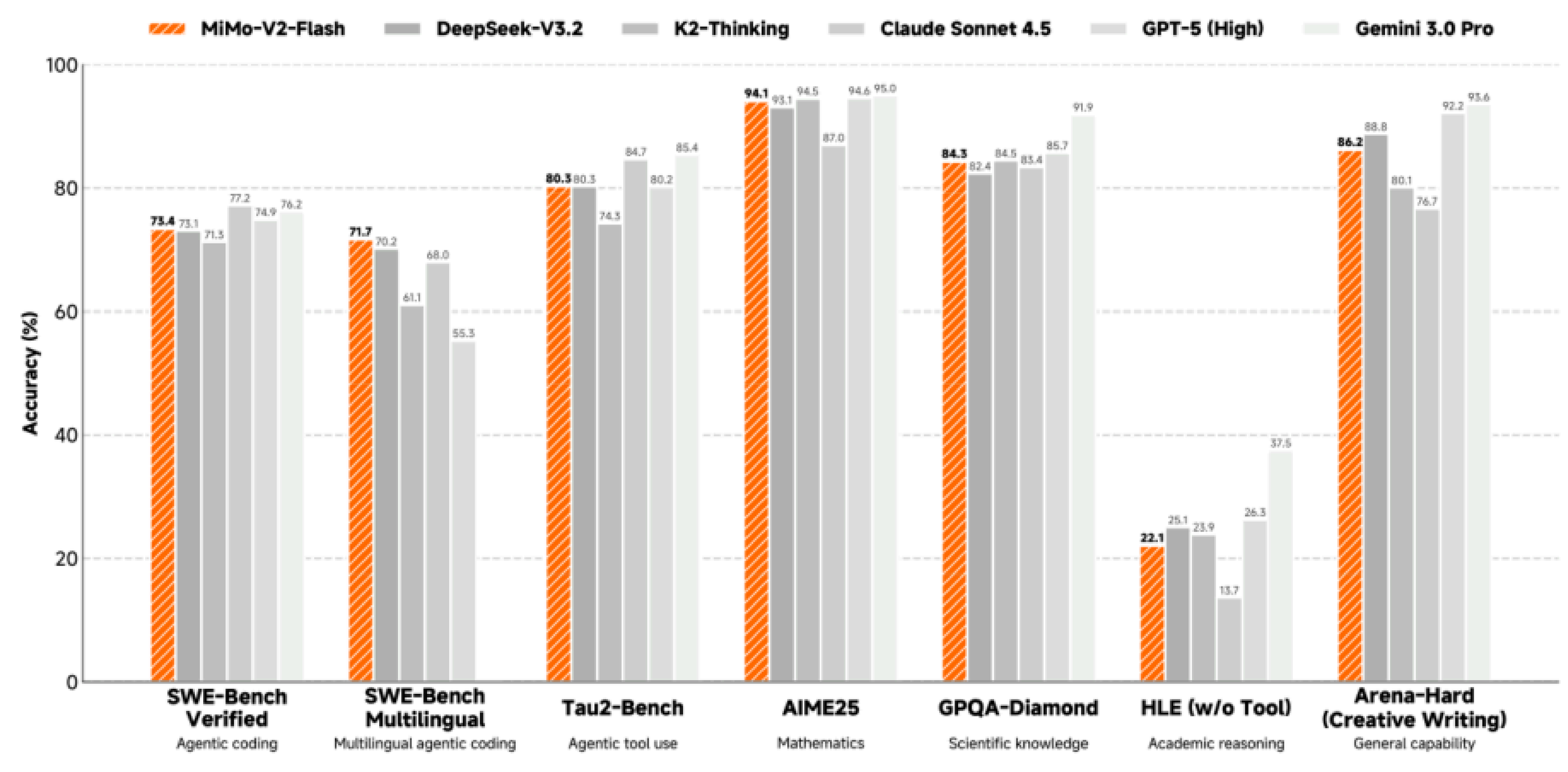


Figure 1 Benchmark performance of MiMo-V2-Flash.

FIGURE 2. An illustration of MiMo-V2-Flash model architecture. The model comprise

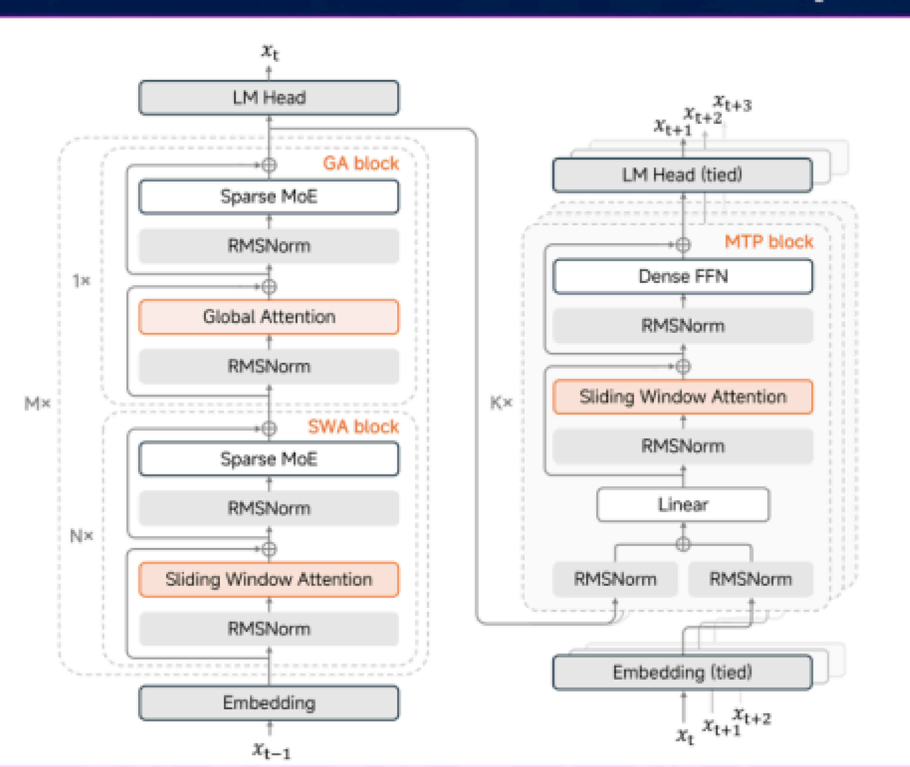
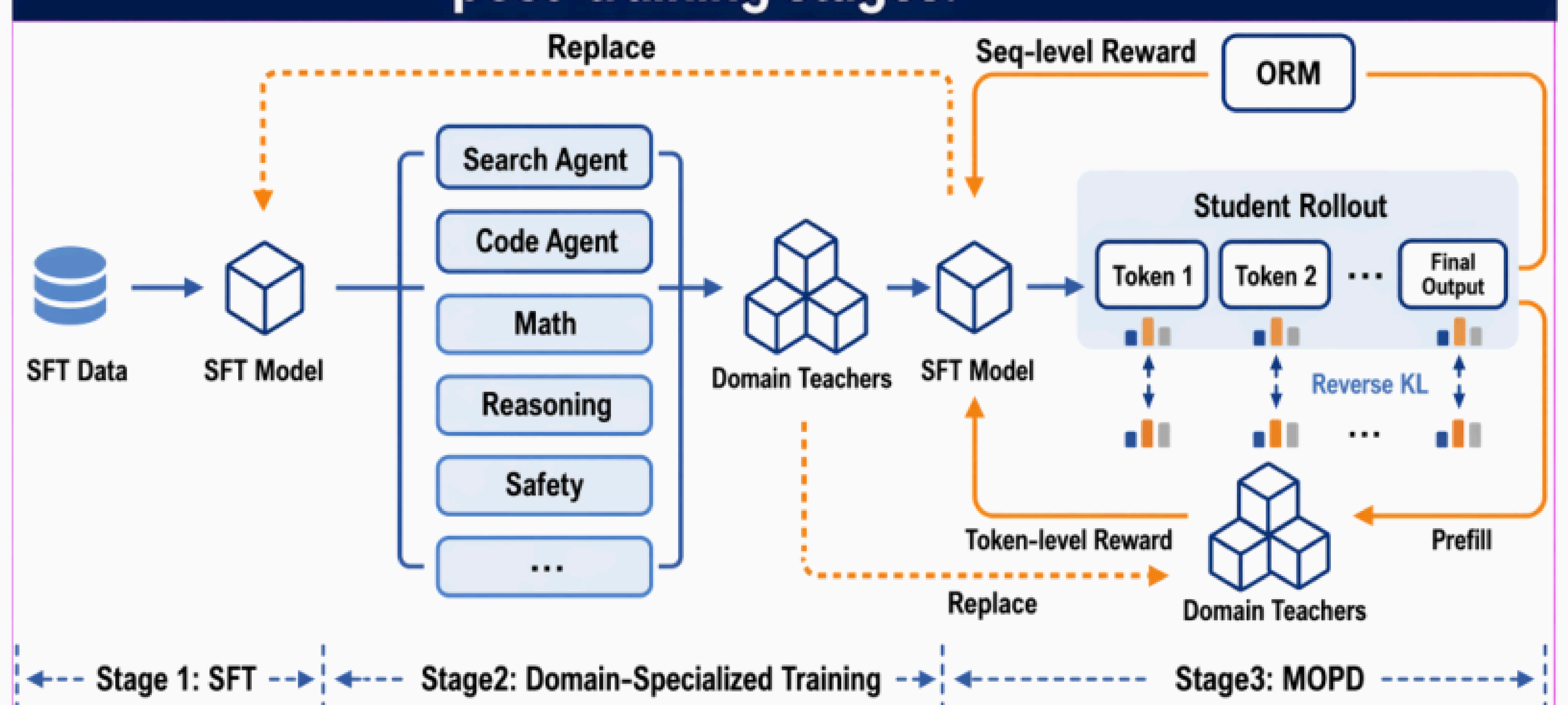


FIGURE 3. Overview of MiMo-V2-Flash post-training stages.



KEY NUMBERS

- **309B** Total model parameters
- **15B** Active parameters per token
- **3.6** Acceptance length in speculative decoding
- **2.6×** Decoding speedup with MTP

RESULTS

73.4% SWE-Bench Verified	**71.7%** SWE-Bench Multilingual	**94.1%** GPQA-Diamond	**86.2%** HLE (w/o Tool)	**80.3%** AIME25	**84.3%** Tau2-Bench Agentic tool use	Up to **3.6** acceptance length and **2.6×** decoding speedup using three MTP layers.	Surpasses larger full-attention models on LongBench V2 and MRCR benchmarks.

POSTER SPECIFICATIONS

NeurIPS 2025 Main Conference (This Version)
Physical Size: 1219 mm (W) × 914 mm (H) | Orientation: Landscape
Material: Paper | Mounting: Push pins on provided boards

Workshops / Mexico City Posters
Physical Size: 24 in (W) × 36 in (H) | Orientation: Portrait
Material: Lightweight paper | Mounting: Provided poster stands